Chapter 9 Notes

Section 1

Notes

Here is why $A \cup B = A \cup (B - A)$.

On the one hand, suppose that $x \in A \cup B$. Then either $x \in A$ or $x \in B$. In the first case, obviously $x \in A \cup (B - A)$. In the second case, if $x \in A$, then we are done (by the first case). Otherwise $x \in B - A$, meaning $x \in A \cup (B - A)$. On the other hand, suppose that $x \in A \cup (B - A)$. Then obviously $x \in A \cup B$.